GATE DA 2027

Database Management & Warehousing

Topic-wise Question Bank

Weighted by what GATE DA 2024–2026 actually asked

115 practice questions + **9** real previous-year questions • **8** topics

MCQ • MSQ • NAT — the three GATE question types

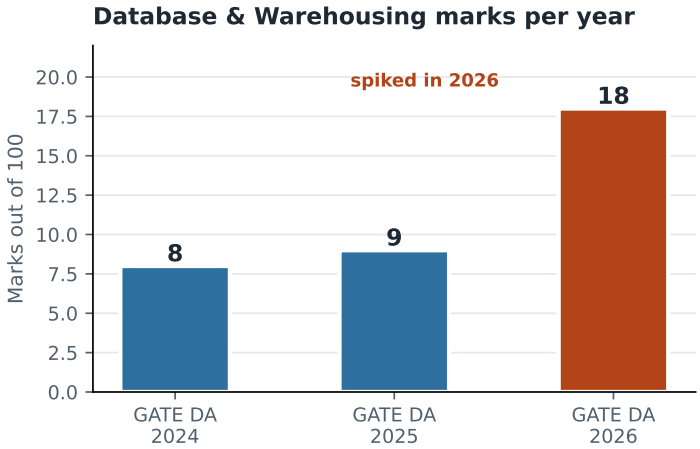
Answer key at the end — answers only, no worked solutions

Every numerical answer in this book (B^+-tree fan-out, FD closures, SQL counts, z-scores) was recomputed independently in Python before the key was fixed — all 16 machine checks agreed. The 9 previous-year questions carry the official GATE answer key. Practice options are shuffled, so the key is not all "A".

Prepared for Ashutosh • August 2026

How this book is built

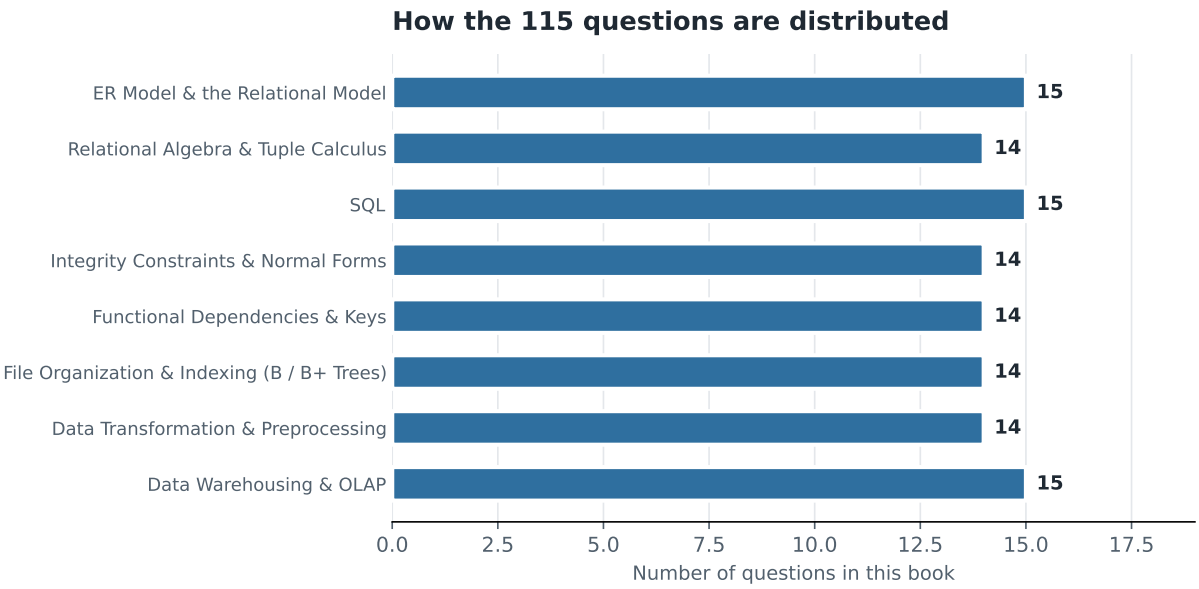
Database & Warehousing is the most *volatile* subject in GATE DA — and that is the whole point. Reading the official papers, it carried only about **8 marks in 2024** and **9 in 2025**, then **jumped to roughly 18 marks in 2026**. A subject that can double overnight is dangerous to skip: the downside of ignoring it (as many did before 2026) is an entire grade band. Plan for **10 to 18 marks** in 2027, not the low figure its early history suggests.

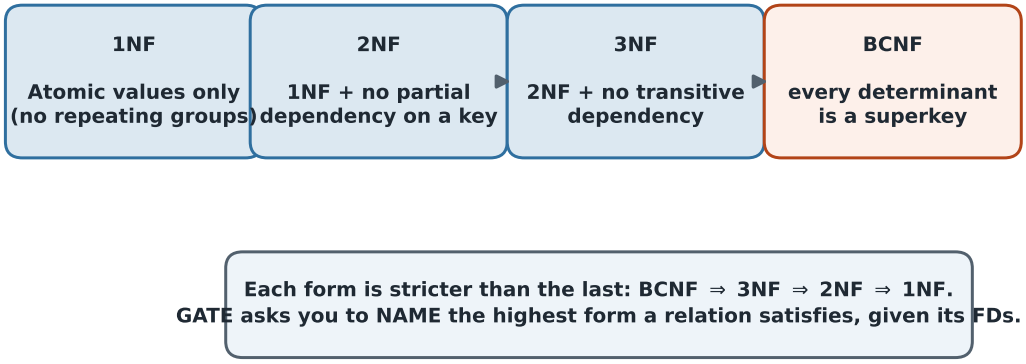


What GATE DA actually asks in Databases

Reading every DB question in the official 2024 and 2026 papers, four habits stand out:

1. **Compute one exact number.** A B^+-tree fan-out from byte sizes (2026 Q.32, answer 195), the rows a correlated SQL subquery returns (2026 Q.60), the tuples a relational-calculus query yields (2026 Q.59), the number of 3NF relations an ER model needs (2026 Q.61). Careful bookkeeping, not memorized theory.
2. **Find the candidate keys from an FD set.** 2026 Q.17 is exactly this. The method never changes: attributes never on a right-hand side are in *every* key; take closures.
3. **Know one clean fact per concept.** Which operation never violates a foreign key (2026 Q.26); which OLAP operation goes Country→State (2026 Q.18, drill-down); B vs B^+ tree.
4. **Data preprocessing sneaks in.** z-score / min–max normalization, binning, sampling (2024 Q.27 is a z-score). These are shared with Machine Learning — learn them once.





Question types and marking (same as the real paper)

Type	Description	Negative marking
MCQ	four options, exactly one correct	$-1/3$ (1-mark) or $-2/3$ (2-mark)
MSQ	four options, one *or more* correct; no partial credit	none
NAT	numerical answer, entered on a virtual keypad	none

This bank is 79% MCQ, 7% MSQ and 14% NAT.

Fifteen facts that answer most DB questions

- Degree = number of columns; cardinality = number of rows; a relation is a *set* of tuples (no duplicates).
- Candidate key = minimal superkey. An attribute never on any FD right-hand side is in *every* candidate key.
- Attribute closure X^+: repeatedly add right sides of FDs whose left side is already in. X is a key iff X^+ = all attributes.
- Superkeys over n attributes with a single-attribute key = 2^{n-1}.
- 1NF: atomic values. 2NF: no partial dependency. 3NF: no transitive dependency ($X \to Y \Rightarrow X$ superkey or Y prime). BCNF: every determinant is a superkey.
- 3NF decomposition is lossless *and* dependency-preserving; BCNF is lossless but may lose dependencies.
- Foreign key: inserting into the referenced table and deleting from the referencing table never violate it; the other two can.
- Relational algebra primitives: $\sigma, \pi, \times, \cup, -, \rho$; $\bowtie$ and $\cap$ are derived. π removes duplicates.
- Cartesian product size = $m \times n$; natural join size $\leq$ that.
- Tuple calculus is declarative; relational algebra is procedural.
- SQL: `WHERE` filters rows, `HAVING` filters groups; `SELECT` keeps duplicates unless `DISTINCT`; `x = NULL` is `UNKNOWN`.
- B^+ tree stores data pointers only in leaves, leaves are linked (good for ranges); a non-leaf with n pointers holds $n - 1$ keys.
- Dense index: one entry per record; sparse: one per block; secondary indexes must be dense.
- Normalization (min–max, z-score), discretization (binning), sampling and compression are the preprocessing operations.
- Warehouse: star (denormalized dims) vs snowflake (normalized dims); OLAP = roll-up, drill-down, slice, dice, pivot.

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Part A — Practice questions by topic

1. ER Model & the Relational Model

15 questions • 12 MCQ, 1 MSQ, 2 NAT • Suggested time: 25 minutes

Q.1 [MCQ • 1 Mark • Moderate]

In the Entity–Relationship (ER) model, a *weak entity* set is one that

(A) is always in third normal form

(B) has no attributes at all

(C) does not have sufficient attributes to form a primary key on its own

(D) participates in no relationships

Q.2 [MCQ • 1 Mark • Easy]

A multivalued attribute in an ER diagram is conventionally drawn as

(A) a diamond (B) a rectangle (C) a double ellipse (D) a dashed ellipse

---

Q.3 [MCQ • 1 Mark • Moderate]

When a multivalued attribute is converted to the relational model, it becomes

(A) a separate relation whose key combines the owner's key and the attribute

(B) a foreign key only

(C) discarded

(D) a single extra column

Q.4 [MCQ • 1 Mark • Moderate]

A many-to-many relationship between two entity sets, when mapped to the relational model, requires

(A) no separate relation

(B) merging the two entity tables

(C) a trigger

(D) a separate relation containing the primary keys of both entities

---

Q.5 [MCQ • 1 Mark • Easy]

The degree of a relation (table) refers to its number of

(A) tuples (rows) (B) keys

(C) foreign keys (D) attributes (columns)

Q.6 [MCQ • 1 Mark • Easy]

The cardinality of a relation refers to its number of

(A) tuples (rows) (B) domains (C) candidate keys (D) attributes

Q.7 [MCQ • 1 Mark • Moderate]

In the relational model, a relation is formally a

(A) list of tuples (duplicates allowed and ordered)

(B) sequence of columns

(C) multiset of ordered rows

(D) set of tuples (so duplicate tuples are not allowed)

Q.8 [MCQ • 1 Mark • Easy]

A `NULL` value in a relational database represents

(A) the number zero

(B) an empty string

(C) a foreign-key error

(D) a missing or unknown value

Q.9 [NAT • 2 Marks • Hard]

Consider an ER model: entities $E_1(A_{11}, A_{12}, A_{13})$ and $E_2(A_{21}, A_{22}, A_{23})$ with A_{22} a multivalued attribute; A_{11}, A_{21} are the primary keys. R_{12} is a many-to-many relationship with total participation of both. The minimum number of relations, each in 3NF, is _____. *(GATE DA 2026, Q.61 pattern.)*

Q.10 [MCQ • 1 Mark • Moderate]

A relationship set in the ER model that involves three entity sets is called

(A) ternary (B) recursive (C) binary (D) unary

Q.11 [MCQ • 1 Mark • Moderate]

Total participation of an entity set E in a relationship R means

(A) R is many-to-many

(B) every entity in E takes part in at least one relationship of R

(C) no entity in E takes part in R

(D) E has a multivalued attribute

Q.12 [MCQ • 1 Mark • Moderate]

A derived attribute in an ER diagram is one whose value

(A) can be computed from other attributes (e.g. age from date of birth)

(B) is multivalued

(C) is a foreign key

(D) is always NULL

Q.13 [MCQ • 1 Mark • Easy]

A *domain* in the relational model is

(A) a primary key

(B) a set of tuples

(C) a relationship

(D) the set of permitted (atomic) values for an attribute

Q.14 [MSQ • 2 Marks • Moderate]

Which are true of a relational *schema*?

(A) It is the structure/description, not the data

(B) An instance of the schema is a set of tuples

(C) It specifies the relation name and its attributes

(D) It changes every time a row is inserted

Q.15 [NAT • 1 Mark • Moderate]

If entity set A has 5 entities and B has 4 entities, the maximum possible number of relationship instances in a many-to-many relationship between them is _____.

2. Relational Algebra & Tuple Calculus

14 questions • 8 MCQ, 1 MSQ, 5 NAT • Suggested time: 25 minutes

Q.1 [MCQ • 1 Mark • Easy]

The relational algebra operation that selects a subset of *rows* satisfying a condition is

(A) selection σ (B) union $\cup$ (C) join $\bowtie$ (D) projection π

Q.2 [MCQ • 1 Mark • Easy]

The operation π (projection) in relational algebra

(A) chooses a subset of rows

(B) computes a Cartesian product

(C) joins two relations

(D) chooses a subset of columns and removes duplicate rows

Q.3 [NAT • 1 Mark • Easy]

Relations R and S have m and n tuples respectively. The Cartesian product $R \times S$ has exactly _____ tuples when $m = 6$, $n = 4$.

Q.4 [MCQ • 1 Mark • Moderate]

The natural join $R \bowtie S$ differs from the Cartesian product in that it

(A) combines only tuples that agree on all common attributes

(B) removes all columns

(C) requires no common attributes

(D) keeps all pairs of tuples

Q.5 [NAT • 2 Marks • Moderate]

If $R(A, B)$ has 3 tuples and $S(B, C)$ has 4 tuples, the natural join $R \bowtie S$ (on B) has *at most* _____ tuples.

Q.6 [MCQ • 1 Mark • Moderate]

The set-difference operation $R - S$ requires that

(A) R and S share no attributes

(B) R and S are union-compatible (same arity and domains)

(C) R has a primary key

(D) R has more tuples than S

Q.7 [MSQ • 2 Marks • Hard]

Which relational-algebra operations form a *complete* set (all others can be derived from them)?

(A) selection σ

(B) projection π

(C) Cartesian product $\times$, union $\cup$ and set difference $-$

(D) aggregation

Q.8 [MCQ • 1 Mark • Moderate]

Tuple relational calculus is a ___ query language, whereas relational algebra is ___.

(A) procedural; declarative

(B) procedural; procedural

(C) declarative; declarative

(D) declarative; procedural

Q.9 [NAT • 2 Marks • Hard]

Given $X(P, Q, R) = \{(P1, Q1, R1), (P2, Q2, R2), (P3, Q3, R2)\}$ and $Y(P, S) = \{(P1, 10), (P1, 15), (P2, 20), (P3, 1)\}$, the tuple calculus query $\{t \mid t \in X \wedge \exists m \in Y(m[P] = t[P] \wedge m[S] > 1)\}$ returns _____ tuples. *(GATE DA 2026, Q.59 pattern.)*

Q.10 [MCQ • 1 Mark • Hard]

The division operation $R \div S$ in relational algebra is used to answer queries of the form

(A) "find tuples related to *some* tuple in S"

(B) "count the tuples in S"

(C) "sort R by S"

(D) "find tuples related to *all* tuples in S"

Q.11 [MCQ • 1 Mark • Easy]

The rename operation ρ is used in relational algebra to

(A) delete a relation

(B) select rows

(C) give a relation or its attributes new names (e.g. for self-joins)

(D) sort a relation

Q.12 [NAT • 1 Mark • Moderate]

R has 10 tuples and S has 7 tuples, and $S \subseteq R$ (union-compatible). The number of tuples in $R \cup S$ is _____.

Q.13 [NAT • 1 Mark • Moderate]

For union-compatible R (10 tuples) and S (7 tuples) with $S \subseteq R$, the number of tuples in $R \cap S$ is _____.

Q.14 [MCQ • 1 Mark • Moderate]

A left outer join $R \bowtie S$ preserves

(A) all tuples of R, padding with NULLs where there is no match in S

(B) only matching tuples

(C) no tuples

(D) all tuples of S only

3. SQL

15 questions • 11 MCQ, 1 MSQ, 3 NAT • Suggested time: 27 minutes

Q.1 [MCQ • 1 Mark • Moderate]

In SQL, which clause is used to filter *groups* produced by `GROUP BY`?

(A) `WHERE` (B) `SELECT` (C) `HAVING` (D) `ORDER BY`

Q.2 [MCQ • 1 Mark • Easy]

The SQL `WHERE` clause filters

(A) columns

(B) the final sort order

(C) groups after aggregation

(D) individual rows before grouping

Q.3 [MCQ • 1 Mark • Moderate]

By default, `SELECT` in SQL

(A) groups the output

(B) sorts the output

(C) keeps duplicate rows (use `DISTINCT` to remove them)

(D) removes duplicate rows

Q.4 [MCQ • 1 Mark • Moderate]

Which SQL aggregate function counts *all* rows in a group, including those with NULLs in some column?

(A) `AVG(*)`

(B) `COUNT(column)`

(C) `SUM(*)`

(D) `COUNT(*)`

Q.5 [MCQ • 1 Mark • Moderate]

A comparison such as `salary = NULL` in SQL evaluates to

(A) `TRUE`

(B) `FALSE`

(C) `UNKNOWN` (use `IS NULL` instead)

(D) a syntax error

Q.6 [NAT • 2 Marks • Hard]

Consider `Account(AccNo, Balance)` with balances $5000, 5000, 10000, 15000, 18000$. The query selects each `AccNo` where the number of accounts with a strictly larger balance is $\geq$ the number with a strictly smaller balance. The number of rows returned is _____. *(GATE DA 2026, Q.60 pattern.)*

Q.7 [NAT • 1 Mark • Easy]

The SQL query `SELECT COUNT(DISTINCT dept) FROM Employee` on a table with departments $\{A, A, B, C, C, C\}$ (6 rows) returns _____.

Q.8 [MCQ • 1 Mark • Easy]

Which SQL join returns only the rows that have matching values in both tables?

(A) `CROSS JOIN` (B) `INNER JOIN`

(C) `LEFT OUTER JOIN` (D) `FULL OUTER JOIN`

Q.9 [MCQ • 1 Mark • Easy]

The clause `ORDER BY salary DESC` sorts the result in

(A) random order (B) increasing order of salary

(C) order of insertion (D) decreasing order of salary

Q.10 [MCQ • 2 Marks • Hard]

In SQL, the logical order in which clauses are *evaluated* is

(A) SELECT → FROM → WHERE → ORDER BY

(B) SELECT → WHERE → FROM → HAVING

(C) WHERE → FROM → SELECT → GROUP BY

(D) FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY

Q.11 [NAT • 1 Mark • Easy]

Table `T` has 4 rows and table `U` has 5 rows. `SELECT * FROM T, U` (a cross join) returns _____ rows.

Q.12 [MSQ • 2 Marks • Moderate]

Which statements about SQL `GROUP BY` are correct?

(A) `WHERE` can reference aggregate results

(B) Aggregate functions operate within each group

(C) Non-aggregated columns in `SELECT` must appear in `GROUP BY`

(D) `HAVING` can reference aggregate results

Q.13 [MCQ • 1 Mark • Moderate]

A *correlated* subquery is one that

(A) is always faster than a join

(B) references a column from the outer query, and is (conceptually) re-evaluated per outer row

(C) can be evaluated once, independently of the outer query

(D) returns no rows

Q.14 [MCQ • 1 Mark • Easy]

To retrieve rows where a column value is one of a given list, SQL provides the operator

(A) `BETWEEN` (B) `LIKE` (C) `EXISTS` (D) `IN`

Q.15 [MCQ • 1 Mark • Moderate]

The `UNION` operator in SQL (as opposed to `UNION ALL`)

(A) keeps all duplicates

(B) removes duplicate rows from the combined result

(C) sorts by the first column only

(D) performs a join

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4. Integrity Constraints & Normal Forms

14 questions • 12 MCQ, 2 MSQ, 0 NAT • Suggested time: 28 minutes

Q.1 [MCQ • 1 Mark • Easy]

A relation is in *first normal form* (1NF) if

(A) every attribute holds only atomic (indivisible) values

(B) it has no transitive dependencies

(C) it has no partial dependencies

(D) every determinant is a candidate key

Q.2 [MCQ • 1 Mark • Moderate]

A relation is in *second normal form* (2NF) if it is in 1NF and

(A) every determinant is a superkey

(B) there are no transitive dependencies

(C) it has a single candidate key

(D) no non-prime attribute is partially dependent on any candidate key

---

Q.3 [MCQ • 1 Mark • Moderate]

A relation is in *third normal form* (3NF) if, for every functional dependency $X \to Y$, at least one of these holds:

(A) Y is a foreign key

(B) X is a superkey, or Y is a prime attribute

(C) X and Y are disjoint

(D) X is a candidate key only

Q.4 [MCQ • 1 Mark • Moderate]

A relation is in *Boyce–Codd normal form* (BCNF) if, for every non-trivial functional dependency $X \to Y$,

(A) Y is a candidate key

(B) Y is a prime attribute

(C) X is a superkey

(D) X is a single attribute

Q.5 [MCQ • 1 Mark • Easy]

Which normal form is the strongest among the following (implies all the others)?

(A) 3NF (B) BCNF (C) 2NF (D) 1NF

Q.6 [MCQ • 1 Mark • Moderate]

A *transitive* dependency (non-prime → non-prime via a non-key attribute) is removed when a relation is brought to

(A) 3NF (B) none of these (C) 2NF (D) 1NF

Q.7 [MCQ • 1 Mark • Moderate]

Referential integrity requires that a foreign-key value in one relation

(A) either matches a primary-key value in the referenced relation or is NULL

(B) is never NULL

(C) is a candidate key

(D) is always unique

Q.8 [MCQ • 1 Mark • Moderate]

Entity integrity requires that

(A) no attribute of a primary key may be NULL

(B) all attributes are unique

(C) every relation has one attribute

(D) every foreign key is NULL

Q.9 [MSQ • 2 Marks • Hard]

Relations r on $R(A, B)$ and s on $S(E, C)$ where A is the primary key of R and E is a foreign key of S referencing A. Which operation(s) will **never** violate the foreign-key constraint? *(GATE DA 2026, Q.26 pattern.)*

(A) Deleting a record from r

(B) Inserting a record into r

(C) Deleting a record from s

(D) Inserting a record into s

Q.10 [MCQ • 2 Marks • Hard]

Decomposing a relation into 3NF can always be done while preserving both

(A) only losslessness, never dependencies

(B) only dependencies, never losslessness

(C) neither property

(D) losslessness and all functional dependencies

Q.11 [MCQ • 2 Marks • Hard]

BCNF decomposition is always lossless but may *not*

(A) reach 1NF

(B) preserve all functional dependencies

(C) remove partial dependencies

(D) be lossless

Q.12 [MCQ • 1 Mark • Moderate]

A *prime attribute* is one that

(A) is a member of some candidate key

(B) is not part of any key

(C) is a foreign key

(D) is always NULL

Q.13 [MSQ • 2 Marks • Moderate]

Which are examples of integrity constraints?

(A) Foreign-key (referential) constraint

(B) Domain constraint (allowed values / type)

(C) An index on a column

(D) Primary-key (uniqueness) constraint

Q.14 [MCQ • 2 Marks • Hard]

Relation $R(A, B, C)$ with only the FD $A \to B$. This relation is in 1NF but violates 2NF if the candidate key is

(A) $\{A, C\}$ (then B is partially dependent on A)

(B) $\{A, B, C\}$

(C) $\{B\}$

(D) $\{A\}$

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5. Functional Dependencies & Keys

14 questions • 11 MCQ, 1 MSQ, 2 NAT • Suggested time: 28 minutes

Q.1 [MCQ • 1 Mark • Easy]

A *superkey* of a relation is a set of attributes that

(A) is always a single attribute

(B) functionally determines all attributes of the relation

(C) has the fewest attributes

(D) contains no prime attribute

Q.2 [MCQ • 1 Mark • Moderate]

A *candidate key* is

(A) a minimal superkey (no proper subset is a superkey)

(B) the chosen primary key only

(C) any superkey

(D) a foreign key

Q.3 [MCQ • 1 Mark • Moderate]

For $R(A, B, C, D)$ with FDs $A \to B$, $B \to C$, $C \to D$, the closure A^+ is

(A) $\{A\}$ (B) $\{A, B\}$ (C) $\{A, B, C, D\}$ (D) $\{A, B, C\}$

Q.4 [MCQ • 1 Mark • Moderate]

For $R(A, B, C, D)$ with FDs $A \to B$, $B \to C$, $C \to D$, the (only) candidate key is

(A) D (B) BC (C) A (D) AB

Q.5 [MCQ • 1 Mark • Easy]

Armstrong's axiom of *transitivity* states that if $X \to Y$ and $Y \to Z$ then

(A) $X \to Z$ (B) $Y \to X$ (C) $Z \to X$ (D) $XZ \to Y$

Q.6 [MSQ • 2 Marks • Moderate]

Which are Armstrong's axioms (sound and complete inference rules for FDs)?

(A) Commutativity: if $X \to Y$ then $Y \to X$

(B) Transitivity: if $X \to Y$ and $Y \to Z$ then $X \to Z$

(C) Augmentation: if $X \to Y$ then $XZ \to YZ$

(D) Reflexivity: if $Y \subseteq X$ then $X \to Y$

Q.7 [MCQ • 2 Marks • Hard]

$R(A, B, C, D, E)$ with FDs $A \to BC$, $CD \to E$, $E \to A$. Which of the following is a candidate key? *(GATE DA 2026, Q.17 pattern.)*

(A) BC (B) CE (C) AB (D) AD

Q.8 [MCQ • 1 Mark • Moderate]

An attribute that appears *only* on the left-hand side of every functional dependency (never on a right-hand side) must

(A) never be part of a key

(B) be part of every candidate key

(C) be a prime attribute only sometimes

(D) be functionally dependent on the key

Q.9 [MCQ • 1 Mark • Moderate]

A functional dependency $X \to Y$ is *trivial* if

(A) X is a key

(B) $Y \subseteq X$

(C) $X \cap Y = \varnothing$

(D) $X \subseteq Y$

Q.10 [NAT • 2 Marks • Hard]

For $R(A, B, C)$ with FDs $AB \to C$ and $C \to A$, the number of candidate keys is _____.

Q.11 [MCQ • 2 Marks • Hard]

A *minimal cover* (canonical cover) of a set of FDs is one in which

(A) the right sides are maximised

(B) every FD has a single attribute on the right, no redundant FDs, and no redundant left-side attributes

(C) all FDs have been removed

(D) every attribute is a key

Q.12 [MCQ • 1 Mark • Moderate]

If $X \to Y$ and $X \to Z$, then by the *union* rule

(A) $Y \to Z$ (B) $X \to Y \cap Z$ (C) $X \to YZ$ (D) $YZ \to X$

Q.13 [NAT • 2 Marks • Hard]

For $R(A, B, C, D)$ with candidate key A, the number of superkeys is _____. (Any set containing A is a superkey.)

Q.14 [MCQ • 1 Mark • Moderate]

The closure of a set of functional dependencies, F^+, is

(A) the set of all attributes

(B) the minimal cover of F

(C) the set of all FDs logically implied by F

(D) the candidate key

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6. File Organization & Indexing (B / B$^+$ Trees)

14 questions • 13 MCQ, 0 MSQ, 1 NAT • Suggested time: 25 minutes

Q.1 [MCQ • 1 Mark • Moderate]

In a B$^+$ tree, the actual data-record pointers are stored

(A) only in the leaf nodes

(B) in every internal node

(C) nowhere

(D) only in the root

Q.2 [MCQ • 1 Mark • Moderate]

A key difference between a B tree and a B$^+$ tree is that in a B$^+$ tree

(A) keys are not sorted

(B) the leaf nodes are linked, supporting efficient range queries

(C) internal nodes store data records

(D) there are no leaf nodes

Q.3 [MCQ • 1 Mark • Moderate]

A B$^+$ tree non-leaf node has n node (tree) pointers. It stores how many search keys?

(A) $2n$ (B) n (C) $n+1$ (D) $n-1$

Q.4 [NAT • 2 Marks • Hard]

B$^+$ tree: node size 4096 bytes, node pointer 10 bytes, key 11 bytes. A non-leaf node has n node pointers and $n-1$ keys, so $10n + 11(n-1) \leq 4096$. The maximum n is _____. *(GATE DA 2026, Q.32 pattern.)*

Q.5 [MCQ • 1 Mark • Moderate]

A *dense* index has

(A) one entry per data block only

(B) no entries

(C) one index entry for every search-key value in the data file

(D) entries only for the primary key

Q.6 [MCQ • 1 Mark • Moderate]

A *sparse* index has

(A) one entry per record

(B) one entry per attribute

(C) no ordering

(D) one index entry per data block (not per record)

Q.7 [MCQ • 1 Mark • Moderate]

A *primary* (clustering) index is built on

(A) the ordering key field of a sequentially ordered file

(B) a hash field

(C) a foreign key (D) any non-ordering field

Q.8 [MCQ • 1 Mark • Moderate]

A B^+ tree of order (max pointers) p has a height that grows as

(A) $O(\sqrt{N})$ (B) $O(N)$

(C) $O(\log_p N)$ for N keys (D) $O(1)$

Q.9 [MCQ • 1 Mark • Moderate]

Which indexing option best speeds up an *equality* search on a single attribute?

(A) a hash index (B) a bitmap on a high-cardinality column

(C) no index (D) a sparse index on a different column

Q.10 [MCQ • 1 Mark • Moderate]

Which indexing structure best supports a *range* query such as `salary BETWEEN 40000 AND 60000`?

(A) a hash index (B) no index (C) a bitmap index (D) a B^+ tree index

Q.11 [MCQ • 2 Marks • Hard]

B^+ tree with at most 2 keys per leaf and at most 3 pointers per non-leaf. After inserting a key that splits a full leaf, the middle key is

(A) discarded (B) placed in a new root only

(C) copied up into the parent (the leaf keys are retained) (D) moved up and removed from the leaf

Q.12 [MCQ • 1 Mark • Moderate]

For a file of N records with a B^+ tree index of fan-out f, the number of block accesses to find one record is approximately

(A) f (B) N (C) N/f (D) $\log_f N$

Q.13 [MCQ • 1 Mark • Moderate]

Hashing can degrade to $O(n)$ lookup time when

(A) many keys collide into the same bucket (B) keys are unique

(C) the hash function is perfect (D) the table is empty

Q.14 [MCQ • 2 Marks • Hard]

A secondary index on a non-ordering field is necessarily

(A) sparse (B) dense (C) clustered (D) a hash index

7. Data Transformation & Preprocessing

14 questions • 10 MCQ, 1 MSQ, 3 NAT • Suggested time: 25 minutes

Q.1 [MCQ • 1 Mark • Moderate]

Min–max normalization rescales a value x to the range $[0, 1]$ using

(A) $\log x$

(B) $\dfrac{x}{\max}$

(C) $\dfrac{x - \mu}{\sigma}$

(D) $\dfrac{x - \min}{\max - \min}$

Q.2 [MCQ • 1 Mark • Moderate]

z-score (standard) normalization transforms x to

(A) $\dfrac{x}{\sigma}$

(B) $x - \mu$

(C) $\dfrac{x - \min}{\max - \min}$

(D) $\dfrac{x - \mu}{\sigma}$

Q.3 [NAT • 2 Marks • Moderate]

Income has mean $\mu = 96000$ and standard deviation $\sigma = 21000$. The z-score of $x = 106000$ (two decimals) is _____. *(GATE DA 2024, Q.27 pattern.)*

Q.4 [NAT • 1 Mark • Moderate]

A value $x = 40$ from a feature with min $= 20$, max $= 100$ is min–max normalized to _____ (two decimals).

Q.5 [MCQ • 1 Mark • Easy]

Discretization of a continuous attribute means

(A) removing it

(B) converting it into a finite number of intervals / categories

(C) taking its logarithm

(D) scaling it to $[0, 1]$

Q.6 [MCQ • 1 Mark • Moderate]

Equal-*width* binning divides the range of a feature into

(A) one interval per point

(B) intervals with equal counts of points

(C) intervals of equal size

(D) intervals chosen by a decision tree

Q.7 [MCQ • 1 Mark • Moderate]

Equal-*frequency* (equal-depth) binning creates bins that

(A) each span the same numeric width

(B) each contain (approximately) the same number of data points

(C) are chosen at random

(D) contain one point each

Q.8 [MCQ • 2 Marks • Moderate]

Values $10, 20, 30, 40, 50, 60$ are split into 3 equal-frequency bins. The second bin contains

(A) $\{10, 20\}$ (B) $\{20, 30\}$ (C) $\{30, 40\}$ (D) $\{40, 50\}$

Q.9 [MCQ • 1 Mark • Easy]

Sampling is used in data preprocessing mainly to

(A) increase the data size

(B) normalize the features

(C) remove all duplicates

(D) work with a representative subset when the full data set is too large

Q.10 [MCQ • 1 Mark • Moderate]

In *stratified* sampling, the population is

(A) sorted then halved

(B) divided into groups (strata) and sampled from each in proportion

(C) never sampled

(D) sampled entirely at random ignoring groups

Q.11 [MCQ • 1 Mark • Moderate]

Lossless data compression is characterized by

(A) it increases the file size

(B) it only works on images

(C) the original data can be reconstructed exactly

(D) some information is always discarded

Q.12 [MSQ • 2 Marks • Moderate]

Which of the following are common data-transformation / preprocessing steps?

(A) Sampling

(B) Normalization (min–max or z-score)

(C) Discretization (binning)

(D) Increasing the learning rate

Q.13 [MCQ • 1 Mark • Moderate]

After z-score normalization, a feature has (approximately)

(A) mean 1 and variance 0

(B) mean 0 and standard deviation 1

(C) minimum 0 and maximum 1

(D) all values in $[-1, 1]$ exactly

Q.14 [NAT • 1 Mark • Easy]

The mean of the values $2, 4, 6, 8$ is 5. After mean-centering (subtracting the mean), the value 8 becomes _____.

8. Data Warehousing & OLAP

15 questions • 14 MCQ, 1 MSQ, 0 NAT • Suggested time: 28 minutes

Q.1 [MCQ • 1 Mark • Easy]

A data warehouse is best described as

(A) a transactional (OLTP) database optimized for many small writes

(B) a subject-oriented, integrated, time-variant, non-volatile collection of data for decision support

(C) a message queue

(D) a single spreadsheet

Q.2 [MCQ • 1 Mark • Moderate]

In a *star schema*, the central fact table is surrounded by

(A) no other tables

(B) index files

(C) other fact tables

(D) denormalized dimension tables

Q.3 [MCQ • 1 Mark • Moderate]

A *snowflake schema* differs from a star schema in that its dimension tables are

(A) removed entirely

(B) turned into fact tables

(C) normalized into multiple related tables

(D) denormalized into one table

Q.4 [MCQ • 1 Mark • Moderate]

In a data warehouse, a *fact table* typically stores

(A) only descriptive text

(B) numeric measures and foreign keys to the dimension tables

(C) only primary keys

(D) the ER diagram

Q.5 [MCQ • 1 Mark • Moderate]

The OLAP operation that moves from a coarser level of aggregation to a finer level of detail (e.g. Country → State) is

(A) slice (B) roll-up (C) drill-down (D) dice

Q.6 [MCQ • 2 Marks • Moderate]

A 3-D data cube shows Sales Quantity by Product Type, Month and Country. To instead view it by Product Type, Month and *State*, the required OLAP operation is *(GATE DA 2026, Q.18 pattern)*

(A) roll-up (B) drill-down (C) dicing (D) slicing

Q.7 [MCQ • 1 Mark • Moderate]

The OLAP operation that aggregates data by climbing up a concept hierarchy (e.g. Month → Year) is

(A) pivot (B) drill-down (C) dice (D) roll-up

Q.8 [MCQ • 1 Mark • Moderate]

The OLAP *slice* operation

(A) rotates the cube

(B) selects a sub-cube on several dimensions

(C) aggregates to the top level

(D) selects a single value for one dimension, reducing the cube's dimensionality by one

Q.9 [MCQ • 1 Mark • Moderate]

The OLAP *dice* operation

(A) fixes exactly one dimension

(B) climbs a concept hierarchy

(C) selects a sub-cube by choosing ranges/values on two or more dimensions

(D) transposes rows and columns

Q.10 [MCQ • 1 Mark • Moderate]

The *pivot* (rotate) OLAP operation

(A) reorients the cube to view the data from a new perspective (swap axes)

(B) deletes a dimension

(C) drills down

(D) aggregates measures

Q.11 [MCQ • 1 Mark • Moderate]

A *concept hierarchy* such as Day < Month < Quarter < Year is used in OLAP to

(A) index the cube (B) normalize the fact table

(C) define the levels for roll-up and drill-down (D) store measures

Q.12 [MCQ • 2 Marks • Hard]

A measure such as `SUM(sales)` that can be aggregated over *all* dimensions is called

(A) additive (B) semi-additive (C) derived (D) non-additive

Q.13 [MSQ • 2 Marks • Moderate]

Which statements about OLTP versus OLAP are correct?

(A) OLAP systems are optimized for high-volume single-row updates

(B) Data warehouses primarily support OLAP

(C) OLTP handles many short read/write transactions

(D) OLAP is optimized for complex analytical queries over large histories

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Q.14 [MCQ • 1 Mark • Moderate]

A data cube with 3 dimensions, each having a concept hierarchy, supports how many OLAP aggregation operations conceptually per hierarchy step? The two that move *along* a hierarchy are

(A) roll-up and drill-down

(B) slice and dice

(C) insert and delete

(D) pivot and slice

Q.15 [MCQ • 2 Marks • Hard]

An *average* measure (e.g. average temperature) that cannot be summed across the time dimension is an example of a ___ measure.

(A) foreign (B) non-additive (C) prime (D) additive

Part B — Previous-year questions (with official keys)

These are **genuine questions from the official GATE DA papers** in your archive. Each carries the **official GATE answer key** and its options are kept in the original paper order (not shuffled). Every question is tagged with its exact year and number. Grouped under the same topics as Part A.

1. ER Model & the Relational Model

Q.1 PYQ [NAT • 2 Marks • Hard]

Consider an ER model with entities $E_1(A_{11}, A_{12}, A_{13})$ and $E_2(A_{21}, A_{22}, A_{23})$, where A_{22} is a multi-valued attribute, and A_{11}, A_{21} are the primary keys of E_1, E_2. R_{12} is a many-to-many relationship between E_1 and E_2, with total participation of both. The minimum number of relations required to convert this ER model into a relational model in which each relation is in 3NF (assuming no other functional dependency) is _____.

Source: GATE DA 2026, Q.61 (2 marks) — official paper & official answer key

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2. Relational Algebra & Tuple Calculus

Q.1 PYQ [NAT • 2 Marks • Hard]

Relation X has columns P, Q, R with rows $(P1, Q1, R1), (P2, Q2, R2), (P3, Q3, R2)$; relation Y has columns P, S with rows $(P1, 10), (P1, 15), (P2, 20), (P3, 1)$. The tuple relational calculus expression $\{t \mid t \in X \wedge \exists z \in X(t[P] = z[P]) \wedge \exists m \in Y(m[P] = t[P] \wedge m[S] > 1)\}$ returns _____ tuples.

Source: GATE DA 2026, Q.59 (2 marks) — official paper (accepted range 2–3)

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3. SQL

Q.1 PYQ [NAT • 2 Marks • Hard]

Relation `Account(AccNo, Balance)` has rows $(A1, 5000), (A2, 5000), (A3, 10000), (A4, 15000), (A5, 18000)$. Consider the query: `SELECT AccNo FROM Account AS A WHERE (SELECT COUNT(*) FROM Account AS B WHERE A.Balance < B.Balance) >= (SELECT COUNT(*) FROM Account AS C WHERE A.Balance > C.Balance)`. The number of rows returned is _____.

Source: GATE DA 2026, Q.60 (2 marks) — official paper & official answer key

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4. Integrity Constraints & Normal Forms

Q.1 PYQ [MSQ • 2 Marks • Moderate]

Two relations r and s are defined on schemas $R(A, B)$ and $S(E, C)$ respectively. A is the primary key of R and E is a foreign key of S referencing A in R. Which of the following operations will NEVER violate the foreign key constraint?

(A) Inserting records into relation r

(B) Deleting records from relation s

(C) Deleting records from relation r

(D) Inserting records into relation s

Source: GATE DA 2026, Q.26 (1 mark) — official paper & official answer key

5. Functional Dependencies & Keys

Q.1 PYQ [MCQ • 1 Mark • Hard]

Let $R(A, B, C, D, E)$ be a relational schema with functional dependency set $F = \{A \rightarrow BC,\ CD \rightarrow E,\ E \rightarrow A\}$. Which of the following statements is correct?

(A) AD, ED and CD are the only candidate keys of R.

(B) AD and ED are the only candidate keys of R.

(C) A, E and CD are the only candidate keys of R.

(D) A and CD are the only candidate keys of R.

Source: GATE DA 2026, Q.17 (1 mark) — official paper & official answer key

6. File Organization & Indexing (B / B$^+$ Trees)

Q.1 PYQ [NAT • 2 Marks • Hard]

A B$^+$ Tree index is built on a key field. Node size = 4096 bytes, Node Pointer size = 10 bytes, Search Key Field size = 11 bytes, Data Record Pointer size = 12 bytes. A Node Pointer points to a sub-tree and a Data Record Pointer points to a block of records. The maximum number of Node Pointers in a non-leaf node of the B$^+$ Tree is _____.

Source: GATE DA 2026, Q.32 (1 mark) — official paper & official answer key

Q.2 PYQ [MCQ • 2 Marks • Hard]

In a B$^+$ Tree the maximum number of key values in each leaf node is 2 and the maximum number of pointers in each non-leaf node is 3. Starting from a given B$^+$ Tree, which option denotes the key value(s) stored in the *root* node after inserting the key value 3? (The given tree has leaves $[1, 2], [5, 7], [8, 9]$ with root keys $\{5, 8\}$ and an internal split occurs.)

(A) 5 (B) 8 (C) 3 and 5 (D) 3, 5 and 8

Source: GATE DA 2026, Q.41 (2 marks) — official paper & official answer key

7. Data Transformation & Preprocessing

Q.1 PYQ [MCQ • 1 Mark • Moderate]

For the attribute income of data scientists, the minimum, maximum, mean and standard deviation are Rs. 46000, Rs. 170000, Rs. 96000 and Rs. 21000 respectively. The z-score normalized value of Rs. 106000 is closest to

(A) 0.217 (B) 0.476 (C) 0.623 (D) 0.481

Source: GATE DA 2024, Q.27 (1 mark) — official paper & official answer key

8. Data Warehousing & OLAP

Q.1 **PYQ** [MCQ • 1 Mark • Moderate]

A 3-dimensional data cube shows Sales Quantity for each combination of Product Type, Month and Country. To instead visualize the Sales Quantity for each combination of Product Type, Month and State, which OLAP operation should be performed?

(A) Slicing (B) Dicing (C) Roll-up (D) Drill-down

Source: GATE DA 2026, Q.18 (1 mark) — official paper & official answer key

Answer Key

Answers only. A key with several letters (e.g. **AB**) is an MSQ, where every listed option must be selected and no other. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted.

Part A — Practice

1. ER Model & the Relational Model

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	C	**3**	A	**4**	D	**5**	D
6	A	**7**	D	**8**	D	**9**	4	**10**	A
11	B	**12**	A	**13**	D	**14**	ABC	**15**	20

2. Relational Algebra & Tuple Calculus

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	**2**	D	**3**	24	**4**	A	**5**	12
6	B	**7**	ABC	**8**	D	**9**	2	**10**	D
11	C	**12**	10	**13**	7	**14**	A		

3. SQL

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	D	**3**	C	**4**	D	**5**	C
6	3	**7**	3	**8**	B	**9**	D	**10**	D
11	20	**12**	BCD	**13**	B	**14**	D	**15**	B

4. Integrity Constraints & Normal Forms

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	**2**	D	**3**	B	**4**	C	**5**	B
6	A	**7**	A	**8**	A	**9**	BC	**10**	D
11	B	**12**	A	**13**	ABD	**14**	A		

5. Functional Dependencies & Keys

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	A	**3**	C	**4**	C	**5**	A
6	BCD	**7**	D	**8**	B	**9**	B	**10**	2
11	B	**12**	C	**13**	8	**14**	C		

6. File Organization & Indexing (B / B^+ Trees)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	**2**	B	**3**	D	**4**	195	**5**	C
6	D	**7**	A	**8**	C	**9**	A	**10**	D
11	C	**12**	D	**13**	A	**14**	B		

7. Data Transformation & Preprocessing

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	D	**3**	0.48	**4**	0.25	**5**	B
6	C	**7**	B	**8**	C	**9**	D	**10**	B
11	C	**12**	ABC	**13**	B	**14**	3		

8. Data Warehousing & OLAP

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	D	**3**	C	**4**	B	**5**	C
6	B	**7**	D	**8**	D	**9**	C	**10**	A
11	C	**12**	A	**13**	BCD	**14**	A	**15**	B

Part B — Previous-year questions (official keys)

Source	Q	Official answer
GATE DA 2026, Q.61 (2 marks)	D1.1	**4**
GATE DA 2026, Q.59 (2 marks)	D2.1	**2**
GATE DA 2026, Q.60 (2 marks)	D3.1	**3**
GATE DA 2026, Q.26 (1 mark)	D4.1	**AB**
GATE DA 2026, Q.17 (1 mark)	D5.1	**A**
GATE DA 2026, Q.32 (1 mark)	D6.1	**195**
GATE DA 2026, Q.41 (2 marks)	D6.2	**A**
GATE DA 2024, Q.27 (1 mark)	D7.1	**B**
GATE DA 2026, Q.18 (1 mark)	D8.1	**D**

Score sheet

Anything below 70% on a topic means the concept, not the practice, is the problem — go back and revise before re-attempting.

Topic	Questions	Correct	Time taken
ER Model & the Relational Model	15		
Relational Algebra & Tuple Calculus	14		
SQL	15		
Integrity Constraints & Normal Forms	14		
Functional Dependencies & Keys	14		
File Organization & Indexing (B / B^+ Trees)	14		
Data Transformation & Preprocessing	14		
Data Warehousing & OLAP	15		
Total	**115**		